# A twelve-variable Diophantine candidate

Strictly positive integer unknowns; exact degree 172; 91 evaluator operations.

The only unknowns are $a, b, c, d, e, f, g, h, i, j, k, \ell \in \mathbb{Z}_{>0}$. Each $\texttt{r01}, \ldots, \texttt{r91}$ below is an abbreviation, not an additional unknown. Evaluate the rows in order using ordinary integer arithmetic; subtraction is integer subtraction, and intermediate values may be negative. The equation is $\boxed{\texttt{r91} = 0}$.

$\texttt{r01} = a \cdot \beta$
$\texttt{r02} = 1 + \texttt{r01}$
$\texttt{r03} = \texttt{r02} \cdot f$
$\texttt{r04} = \texttt{r02} \cdot \texttt{r02}$
$\texttt{r05} = \texttt{r02} \cdot \texttt{r04}$
$\texttt{r06} = \texttt{r05} \cdot g$
$\texttt{r07} = \texttt{r06} \cdot \texttt{r04}$
$\texttt{r08} = \texttt{r02} - b$
$\texttt{r09} = \texttt{r08} - c$
$\texttt{r10} = \texttt{r03} \cdot \texttt{r06}$
$\texttt{r11} = i + j$
$\texttt{r12} = \texttt{r06} \cdot \texttt{r11}$
$\texttt{r13} = i + \texttt{r12}$
$\texttt{r14} = \texttt{r13} \cdot \texttt{r13}$
$\texttt{r15} = \texttt{r10} + \texttt{r06}$
$\texttt{r16} = 4 \cdot \texttt{r15}$
$\texttt{r17} = 3 + \texttt{r16}$
$\texttt{r18} = \texttt{r15} \cdot \texttt{r15}$
$\texttt{r19} = \texttt{r17} + \texttt{r18}$
$\texttt{r20} = \texttt{r15} \cdot \texttt{r13}$
$\texttt{r21} = \texttt{r03} + \texttt{r20}$
$\texttt{r22} = \texttt{r10} \cdot \texttt{r12}$
$\texttt{r23} = \kappa + f$
$\texttt{r24} = \texttt{r09} \cdot \texttt{r23}$
$\texttt{r25} = \texttt{r24} - b$
$\texttt{r26} = \texttt{r02} \cdot \texttt{r25}$
$\texttt{r27} = \texttt{r22} + \texttt{r11}$
$\texttt{r28} = \texttt{r27} + h$
$\texttt{r29} = \texttt{r11} \cdot \texttt{r26}$
$\texttt{r30} = \texttt{r28} - \texttt{r29}$
$\texttt{r31} = \texttt{r30} + g$
$\texttt{r32} = \texttt{r01} \cdot \texttt{r31}$
$\texttt{r33} = \texttt{r32} - \texttt{r26}$
$\texttt{r34} = \texttt{r33} \cdot \texttt{r33}$
$\texttt{r35} = \texttt{r01} \cdot \texttt{r01}$
$\texttt{r36} = \texttt{r22} \cdot \texttt{r27}$
$\texttt{r37} = 1 + \texttt{r36}$
$\texttt{r38} = \texttt{r35} \cdot \texttt{r37}$
$\texttt{r39} = \texttt{r34} - \texttt{r38}$
$\texttt{r40} = \texttt{r08} \cdot \texttt{r01}$
$\texttt{r41} = c + \texttt{r40}$
$\texttt{r42} = \texttt{r04} - 1$
$\texttt{r43} = \texttt{r41} \cdot \texttt{r42}$
$\texttt{r44} = \nu \cdot \texttt{r02}$
$\texttt{r45} = \mu + \texttt{r44}$
$\texttt{r46} = a \cdot \texttt{r45}$
$\texttt{r47} = \texttt{r43} + \texttt{r46}$
$\texttt{r48} = \texttt{r11} - \texttt{r47}$
$\texttt{r49} = \texttt{r48} - 1$
$\texttt{r50} = \ell + f$
$\texttt{r51} = \texttt{r07} \cdot \texttt{r50}$
$\texttt{r52} = \texttt{r51} - \texttt{r49}$
$\texttt{r53} = \texttt{r07} \cdot \texttt{r21}$
$\texttt{r54} = \texttt{r17} \cdot \texttt{r52}$
$\texttt{r55} = \texttt{r53} + \texttt{r54}$
$\texttt{r56} = \texttt{r55} \cdot \texttt{r55}$
$\texttt{r57} = \texttt{r07} \cdot \texttt{r07}$
$\texttt{r58} = \texttt{r19} \cdot \texttt{r14}$
$\texttt{r59} = 1 + \texttt{r58}$
$\texttt{r60} = \texttt{r57} \cdot \texttt{r59}$
$\texttt{r61} = \texttt{r56} - \texttt{r60}$
$\texttt{r62} = 1 + d$
$\texttt{r63} = \texttt{r62} \cdot \texttt{r62}$
$\texttt{r64} = \texttt{r63} - 1$
$\texttt{r65} = \texttt{r19} \cdot \texttt{r64}$
$\texttt{r66} = 1 - \texttt{r65}$
$\texttt{r67} = \texttt{r14} \cdot \texttt{r14}$
$\texttt{r68} = e \cdot \texttt{r13}$
$\texttt{r69} = \texttt{r68} - \texttt{r47}$
$\texttt{r70} = \texttt{r63} \cdot \texttt{r69}$
$\texttt{r71} = \texttt{r70} - \texttt{r13}$
$\texttt{r72} = \texttt{r71} + k$
$\texttt{r73} = \texttt{r72} \cdot \texttt{r72}$
$\texttt{r74} = \texttt{r71} \cdot \texttt{r71}$
$\texttt{r75} = \texttt{r65} \cdot \texttt{r74}$
$\texttt{r76} = \texttt{r75} - 1$
$\texttt{r77} = \texttt{r66} \cdot \texttt{r73}$
$\texttt{r78} = \texttt{r77} + \texttt{r76}$
$\texttt{r79} = \texttt{r65} \cdot \texttt{r66}$
$\texttt{r80} = \texttt{r67} \cdot \texttt{r78}$
$\texttt{r81} = \texttt{r80} - \texttt{r79}$
$\texttt{r82} = \texttt{r81} \cdot \texttt{r81}$
$\texttt{r83} = \texttt{r67} \cdot \texttt{r79}$
$\texttt{r84} = 4 \cdot \texttt{r83}$
$\texttt{r85} = \texttt{r84} \cdot \texttt{r76}$
$\texttt{r86} = \texttt{r82} + \texttt{r85}$
$\texttt{r87} = \texttt{r39} \cdot \texttt{r39}$
$\texttt{r88} = \texttt{r61} \cdot \texttt{r61}$
$\texttt{r89} = \texttt{r86} \cdot \texttt{r86}$
$\texttt{r90} = \texttt{r87} + \texttt{r88}$
$\texttt{r91} = \texttt{r90} + \texttt{r89}$

**The four fixed integers.** All indices in the following formulas are locally bound indices for constants, not unknowns. Set $\rho = 2^{67}$ and $\beta = \rho^{8807} - 1$. Let the fixed set and ordered list be

$$\begin{aligned}\mathcal{S} = \big(&\{0,\ldots,942\} \cup \{983,\ldots,992\} \cup \{1003,\ldots,1011\} \cup \{1013,\ldots,1031\} \\ &\cup \{1063, 1064, 1095, 2195, 3296\}\big) \setminus \{1\},\end{aligned}$$

$$\begin{aligned}\lambda = (&1023, 1024, 1025, 1026, 1027, 1028, 1029, 1030, 1013, 1014, \\ &1015, 1016, 1017, 1018, 1019, 1020, 1021, 1003, 1004, 1005, \\ &1006, 1007, 1008, 1009, 1010, 983, 984, 985, 986, 987, \\ &988, 989, 990, 991, 1031, 1022, 1011, 992, 1095, 2195, \\ &3296).\end{aligned}$$

The list $\lambda$ has 41 entries, numbered from zero. The fixed list $\sigma$ has 943 entries, also numbered from zero, and is printed in decimal on the following pages. Then

$$\mu = \beta - \sum_{u\in\mathcal{S}} \rho^u - 2\rho^{1063},$$

$$\begin{aligned}\nu = \beta + 4 + &\sum_{u=1013}^{1022} \rho^u + \sum_{u=1065}^{1073} \rho^u + \rho^{2195} + \rho^{3296} \\ &+ (2^{51}-2)(\rho^{4393} + \rho^{5495}) + (2^{65}-1)\sum_{u=8792}^{8804} \rho^u + (2^{19}-1)\rho^{8805},\end{aligned}$$

$$\begin{aligned}\kappa = \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ + \sum_{u=0}^{942} \sigma_u(\rho^{4393-u} + \rho^{5495-u}) + \sum_{u=0}^{40} 2^{10+u}(\rho^{4393-\lambda_u} + \rho^{5495-\lambda_u}).\end{aligned}$$

**Decimal selector list.** Read from left to right, top to bottom; each 14-digit block is one decimal integer. Leading zeroes do not change its value. There are exactly 943 blocks.

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